Zero-bias transport and extremal Bernoulli arrays

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Abstract

For a finite sum of independent centered real variables, the Wasserstein distance to its zero-biased law is at most the sum of the third absolute moments divided by twice the total variance. We prove that equality holds exactly when all nonconstant summands have one identical two-point law. The proof treats arbitrary summand distributions: a same-variance Choquet representation reduces the single-law equality problem to three-point laws, and an exact conditional-variance identity resolves the remaining Bernoulli mixtures. A sharp comparison between lattice span, skewness, and zero-bias transport then identifies equality in the Berry–Esseen limit obtained by independently repeating a fixed finite block. The limiting ratio is at most Esseen's constant, with equality exactly for identically distributed Esseen two-point summands and any zero summands.

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1 The problem and the main results

The classical extremal example for the Berry–Esseen inequality is a sum of identically distributed two-point variables. For general independent summands, two separate questions arise: what forces two-point support, and what forces the resulting laws to agree? This paper answers both questions

for equality in the optimal zero-bias transport bound. It then uses that equality classification in a fixed-block Berry–Esseen limit.

For probability laws with finite first moments, write

$$W_1(\mu,\nu) = \inf\{\mathbb{E}|U-V| : \mathcal{L}(U)=\mu,\ \mathcal{L}(V)=\nu\}.$$

We also write $W_1(U,V)$ for the distance between their laws. If X is centered and has variance $v>0$, its zero-biased law is characterized by

$$\mathbb{E}[Xf(X)] = v\mathbb{E}f'(X^*). \tag{1.1}$$

This identity holds for every absolutely continuous f for which the expectations exist. Goldstein and Reinert introduced this transformation and its independent-sum replacement construction [5, Definition 1.1 and Lemma 2.1]. Goldstein and Tyurin independently proved the sharp bound

$$W_1(X,X^*) \le \frac{\mathbb{E}|X|^3}{2v} \tag{1.2}$$

for every centered law with finite third absolute moment [4, Lemma 1.1][6, Theorem 3]. Every nonconstant two-point law attains equality. Our first result characterizes all equality cases, both for one law and for an independent array.

Let $X_1,\dots,X_n$ be independent centered real variables and put

$$S=\sum_{i=1}^n X_i, \qquad V=\sum_{i=1}^n \mathbb{E}X_i^2>0, \qquad R=\sum_{i=1}^n \mathbb{E}|X_i|^3<\infty. \tag{1.3}$$

The denominator R records the original summands. It is generally different from $\mathbb{E}|S|^3$.

Theorem 1.1 (Equality in zero-bias transport)**.** *For the array in* (1.3)*,*

$$2VW_1(S,S^*) \le R. \tag{1.4}$$

Equality holds if and only if all nonconstant coordinates have one identical centered two-point law. Any remaining coordinates are identically zero. In particular, equality in (1.2) *holds if and only if X has a two-point law.*

The distinction between R and $\mathbb{E}|S|^3$ is necessary even for the equality statement. A sum of two or more nonconstant iid Bernoulli variables has more than two support points. Its single-law bound is therefore strict, although its bound with the original third-moment sum is an equality.

For changing arrays with Lyapunov ratio tending to zero and Berry–Esseen ratio tending to Esseen's constant, a companion paper [1] proves that (1.4) approaches equality. Full finite-horizon global optimality in that regime gives the exact equality class. The fixed-block theorem below has its own asymptotic hypotheses.

The second result concerns repetitions of one fixed independent block. Let $S_1,S_2,\dots$ be independent copies of the sum S in (1.3). For each integer $m\ge 1$, define

$$C_m = \frac{\sqrt{m}\,V^{3/2}}{R}\sup_x\left|\mathbb{P}\left\{\frac{S_1+\cdots+S_m}{\sqrt{mV}}\le x\right\}-\Phi(x)\right|. \tag{1.5}$$

This is the Berry–Esseen ratio for the mn original coordinates, whose total variance is mV and whose third-moment sum is mR. Set

$$C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}}, \qquad p_{\mathrm{E}} = \frac{4-\sqrt{10}}{2}.$$

Theorem 1.2 (A fixed block repeated independently)**.** *The limit of* (1.5) *exists and satisfies*

$$\lim_{m\to\infty} C_m \le C_{\mathrm{E}} \frac{2VW_1(S,S^*)}{R} \le C_{\mathrm{E}}. \tag{1.6}$$

Equality with C_{E} *holds if and only if, for some* $h>0$*, every nonconstant coordinate has the law of* $h(B-p_{\mathrm{E}})$*, where* $B\sim\mathrm{Bern}(p_{\mathrm{E}})$*, or every such coordinate has its reflection. The other coordinates are zero.*

Esseen's fixed-law asymptotic formula and sharp moment inequality already give the constant C_{E} for iid sums [3, pp. 160–163]. Theorem 1.2 retains the separate third moments of the original block and identifies equality through transport. The proof depends on a lattice inequality in which entire binomial laws, rather than only two-point laws, attain equality. The original denominator then determines which binomial decompositions attain the Berry–Esseen constant.

The block is fixed throughout the limit. The theorem gives no uniform remainder estimate for changing laws, and it does not prove the sharp Berry–Esseen bound at a finite sample size. Its proof uses the classical asymptotic formula.

2 Zero bias and the independent-sum bound

We use two formulas from Goldstein and Reinert [5, Lemma 2.1(2), (4), and (5)]. The zero-biased law of a centered variable X of variance $v>0$ has density

$$d_X(u) = \frac{1}{v}\mathbb{E}[X\mathbf{1}_{\{X>u\}}]. \tag{2.1}$$

For $u<0$, centering rewrites the numerator as $-\mathbb{E}[X\mathbf{1}_{\{X\le u\}}]$, so the density is nonnegative on both half-lines. The defining identity gives

$$\mathbb{E}X^* = \frac{\mathbb{E}X^3}{2v}, \qquad \mathbb{E}|X^*| = \frac{\mathbb{E}|X|^3}{2v}. \tag{2.2}$$

The first follows by taking $f(x)=x^2/2$; the second follows by taking $f(x)=x|x|/2$.

For laws with finite first moments, the Wasserstein distance is the integral of the absolute CDF difference:

$$W_1(\mu,\nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)|\,dx. \tag{2.3}$$

For example, Goldstein states the optimal quantile coupling in [4, equations (15)–(16)]. This formula makes both mixture convexity and convolution contraction immediate by the triangle inequality and Tonelli's theorem.

Discard the zero coordinates of (1.3) and write $v_i=\mathbb{E}X_i^2$, $\rho_i=\mathbb{E}|X_i|^3$, and $T_i=S-X_i$. Choose an index I, independently of all variables, with $\mathbb{P}(I=i)=v_i/V$, and replace X_I by an independent variable with its zero-biased law. Then

$$S^* \overset{d}{=} T_I + X_I^*. \tag{2.4}$$

Indeed, applying (1.1) to each summand, conditional on the others, gives $\mathbb{E}[Sf(S)]=\sum_i v_i\mathbb{E}f'(T_i+X_i^*)$. Consequently

$$\begin{aligned} W_1(S,S^*) &\le \sum_i \frac{v_i}{V} W_1(S, T_i+X_i^*) \\ &\le \sum_i \frac{v_i}{V} W_1(X_i, X_i^*) \le \frac{R}{2V}. \end{aligned} \tag{2.5}$$

The last step is (1.2). Tyurin's Corollary 1 gives the same independent-sum bound [6].

There are three possible sources of strictness in (2.5): a strict single-coordinate bound, cancellation under convolution, and cancellation between replacement components. The next two sections determine their equality cases.

3 Equality for an arbitrary single law

For a centered law μ of variance $v > 0$ and third absolute moment ρ, put

$$B(\mu) = \frac{2vW_1(\mu, \mu^*)}{\rho}.$$

This quantity is invariant under nonzero scaling and reflection. We first settle laws with at most three atoms, then use an exact mixture at fixed variance to treat every law.

3.1 Two-point laws and strict three-point bounds

A centered two-point law can be written as $h(B - p)$, with $h > 0$, $0 < p < 1$, $q = 1 - p$, and $B \sim \mathrm{Bern}(p)$. Its variance and third absolute moment are

$$v = h^2pq, \qquad \rho = h^3pq(p^2 + q^2).$$

Its zero-biased law is uniform on $[-hp, hq]$, and direct integration gives

$$W_1(\mu, \mu^*) = \frac{h}{2}(p^2 + q^2). \tag{3.1}$$

These are Goldstein's exact two-point formulas [4, Lemma 1.2]; they give $B(\mu) = 1$.

Lemma 3.1. *Every centered law with three distinct support points of positive mass has $B(\mu) < 1$.*

Proof. First suppose none of its atoms is zero. Reflect the law, if necessary, so that its support is $x < y < 0 < z$. Let μ_1 and μ_0 be the centered two-point laws on $\{x, z\}$ and $\{y, z\}$. There is a unique $\alpha \in (0, 1)$ such that

$$\mu = \alpha\mu_1 + (1 - \alpha)\mu_0.$$

The variances are $-xz$ and $-yz$. Thus

$$\mu^* = \beta\mu_1^* + (1 - \beta)\mu_0^*, \qquad \beta = \frac{\alpha x}{\alpha x + (1 - \alpha)y} > \alpha.$$

This variance reweighting is Goldstein's mixture identity [4, Theorem 2.1].

Write $D_j = W_1(\mu_j, \mu_j^*)$ and $D_{10} = W_1(\mu_0, \mu_1^*)$. A coupling formed with weights α, $1 - \beta$, and $\beta - \alpha$ gives

$$W_1(\mu, \mu^*) \le \alpha D_1 + (1 - \beta)D_0 + (\beta - \alpha)D_{10}. \tag{3.2}$$

In contrast, the two-point equalities give

$$\frac{\rho}{2v} = \beta D_1 + (1 - \beta)D_0.$$

The cross-transport comparison in Goldstein's Lemma 3.1 is strict for $x < y < 0 < z$. More explicitly, integrating the two affine CDF pieces, as in [4, equation (28) and the following case], gives

$$D_1 - D_{10} = \begin{cases} \dfrac{-y(y-x)\{y^2+2z^2-y(x+2z)\}}{(z-x)(z-y)^2}, & y(x+z) \le y^2+z^2, \\ \dfrac{z^2}{z-x}, & y(x+z) > y^2+z^2. \end{cases}$$

In the first case the expression in braces is at least $z(z-y) > 0$. The second expression is positive as well. Subtracting (3.2) therefore gives a strictly positive deficit $\rho/(2v) - W_1(\mu, \mu^*)$.

It remains to treat an atom at zero. Write the support as $\{-ph, 0, qh\}$, where $p+q=1$ and, after reflection, $0 < p \le q$. If the middle mass is $1-\tau \in (0,1)$, the endpoint masses are τq and τp. The zero-biased law is uniform on the full interval. Integration on its two sides gives, with $r = p^2+q^2$,

$$B(\mu) = \begin{cases} (q - p + 2\tau^2 p^2)/r, & \tau q \ge p, \\ 1 + 2\tau^2 - 4\tau pq/r, & \tau q < p. \end{cases} \tag{3.3}$$

The deficits are respectively $2p^2(1-\tau^2)/r$ and $2\tau(2pq/r - \tau)$. Both are positive: in the second case, $2pq/r \ge p/q > \tau$. This direct calculation provides strictness at the boundary which Goldstein's proof of Theorem 3.1 treats by continuity. □

3.2 An exact mixture at fixed variance

Normalize the variance of μ to one. Consider the convex set

$$\mathcal{K} = \left\{ \nu : \nu \text{ is a probability law on } \mathbb{R},\ \int x\,\nu(dx) = 0,\ \int x^2\,\nu(dx) \le 1 \right\}.$$

The inequality on the second moment makes this set weakly compact. Indeed, it gives tightness and uniform integrability of first moments, since

$$\int_{|x|>A} |x|\,\nu(dx) \le \frac{1}{A}.$$

Centering is therefore preserved under weak limits, and lower semicontinuity preserves the second-moment bound. The weak topology on $\mathcal{K}$ is metrizable. Embed finite signed measures in the locally convex Hausdorff space whose topology is induced by integration against $C_b(\mathbb{R})$. Choquet's representation theorem applies to $\mathcal{K}$ [2, pp. 333–337, especially Corollary 2 and Lemma 5]. It gives a probability measure π carried by the extreme points, with barycenter μ.

Every extreme point of $\mathcal{K}$ has at most three atoms. To see this, suppose ν charges four disjoint Borel sets $A_1, \dots, A_4$. Choose a nonzero vector $(c_j)_{j=1}^4$ in the nullspace of their three feature rows

$$\left(\int_{A_j} 1\,d\nu \right)_j, \quad \left(\int_{A_j} x\,d\nu \right)_j, \quad \left(\int_{A_j} x^2\,d\nu \right)_j.$$

The bounded function $g = \sum_j c_j \mathbf{1}_{A_j}$ is nonzero ν-almost surely on a set of positive mass. For sufficiently small $\varepsilon > 0$, the distinct measures $(1 \pm \varepsilon g)\nu$ belong to $\mathcal{K}$ and have average ν, contradicting extremality. A law not supported on at most three points admits such four sets.

The barycenter identity first holds for bounded continuous functions. It identifies μ with the mixture measure $\int \nu\,\pi(d\nu)$, and hence extends to nonnegative Borel functions. In particular,

$$1 = \int \left(\int x^2\,\nu(dx) \right) \pi(d\nu), \qquad \rho(\mu) = \int \rho(\nu)\,\pi(d\nu). \tag{3.4}$$

Every inner second moment is at most one. Therefore it equals one for π-almost every component. The decomposition is consequently into centered laws of the *same* variance, with no degenerate component.

For these components the zero-bias transformation uses the same mixture weights. One can check this without any choice of coupled variables: for a Borel set A, the measurable positive kernel

$$\nu^*(A) = \int x \left(\int_0^x \mathbf{1}_A(u)\, du \right) \nu(dx) \tag{3.5}$$

has total mass one and satisfies (1.1). For negative x, the inner integral is oriented, so its product with x remains nonnegative. Tonelli's theorem gives

$$\mu^* = \int \nu^*\, \pi(d\nu). \tag{3.6}$$

All required first moments are integrable by (2.2) and (3.4). Thus

$$W_1(\mu, \mu^*) \le \int W_1(\nu, \nu^*)\, \pi(d\nu) \le \frac{1}{2} \int \rho(\nu)\, \pi(d\nu) = \frac{\rho(\mu)}{2}. \tag{3.7}$$

Goldstein's Theorem 2.2 gives this convexity at fixed variance; it also explains why unequal mixture variances cannot be used here [4, Remark 2.2]. The exact barycenter above permits its use in the equality problem for an arbitrary law.

If $B(\mu) = 1$, the nonnegative component deficits in (3.7) have integral zero. Lemma 3.1 then shows that almost every component is a two-point law.

3.3 Distinct binary components cancel

A centered unit-variance two-point law is uniquely parametrized by $a > 0$:

$$\nu_a = \frac{1}{1+a^2}\delta_{-a} + \frac{a^2}{1+a^2}\delta_{1/a}.$$

Its signed third moment is $c(a) = 1/a - a$, a strictly decreasing continuous function with a continuous inverse. This parametrization is Borel, since moments are measurable limits of bounded truncations. Push the representing measure π to the parameter a.

The signed CDF difference from the uniform zero-biased law is

$$f_a(x) = F_{\nu_a}(x) - F_{\nu_a^*}(x) = \frac{1}{1+a^2} - \frac{x+a}{a+1/a} \quad (-a < x < 1/a), \tag{3.8}$$

and is zero outside that interval, apart from irrelevant endpoint values. Its zero is $c(a)$. If $a < b$, then $f_a > 0$ and $f_b < 0$ on

$$(\max\{-a, c(b)\},\ \min\{1/b, c(a)\}). \tag{3.9}$$

This interval has positive length: each of $-a$ and $c(b)$ is strictly smaller than each of $1/b$ and $c(a)$.

If the parameter distribution is not a point mass, choose two distinct points $a < b$ in its support. On a compact subinterval of (3.9), sufficiently small parameter neighborhoods retain the two opposite signs, bounded away from zero. Both neighborhoods have positive probability. For every x in that subinterval,

$$\mathbb{E}|f_a(x)| - |\mathbb{E} f_a(x)| = 2\min\{\mathbb{E}(f_a(x))_+, \mathbb{E}(f_a(x))_-\} > 0.$$

Here $u_+ = \max(u, 0)$ and $u_- = \max(-u, 0)$. Integrating gives a strict triangle inequality in the first line of (3.7). Fubini is justified by $\int \mathbb{E}|f_a(x)|\, dx = \int \rho(\nu_a)\, \pi(da)/2 < \infty$. Therefore equality forces a single binary component. Scaling back proves the single-law assertion in Theorem 1.1.

4 Equality for independent arrays

If any nonconstant coordinate is not two-point, the last inequality in (2.5) is strict. We may therefore write every nonconstant coordinate as

$$X_i = h_i(B_i - p_i), \qquad h_i > 0, \quad 0 < p_i < 1,$$

with independent $B_i \sim \mathrm{Bern}(p_i)$. Two remaining calculations determine the gaps and probabilities separately.

4.1 Unequal support diameters give strict convolution loss

Suppose a coordinate has the largest support diameter h and another coordinate has diameter $0 < \delta < h$. In raw coordinates the signed CDF difference between the first coordinate and its uniform zero-biased law is

$$f(x) = q - x/h \quad (0 < x < h), \qquad f(x) = 0 \text{ otherwise},$$

where q is its lower-atom probability. The deleted sum has two atoms of positive probability separated by δ: fix all its Bernoulli coordinates at their lower atoms, and then change only the coordinate of diameter δ.

The two corresponding translates of f have opposite signs on a translate of

$$(\max\{qh, \delta\},\ \min\{qh + \delta, h\}).$$

Its length is positive because $0 < q < 1$ and $0 < \delta < h$. Consequently $|\mathbb{E}f(x - T)| < \mathbb{E}|f(x - T)|$ on a set of positive measure. After integration, convolution contraction is strict for the largest-gap coordinate in (2.5). Equality in (1.4) therefore forces one common diameter h.

4.2 An exact deficit for a common diameter

Assume there are n nonconstant coordinates, all of diameter h. Put

$$K = \sum_i B_i, \qquad \mu = \sum_i p_i, \qquad w_i = \frac{h^2 p_i(1 - p_i)}{V}, \qquad c_{i,k} = \mathbb{P}\left\{\sum_{j \neq i} B_j = k\right\}.$$

For $0 \le k \le n - 1$, define

$$b_k = \sum_i w_i c_{i,k}, \qquad \pi_{i,k} = \frac{w_i c_{i,k}}{b_k}, \qquad Q_k = \sum_i \pi_{i,k}(1 - p_i). \tag{4.1}$$

Every b_k is positive, and $\sum_k b_k = 1$.

Proposition 4.1 (Conditional-variance identity)**.** *For a common-diameter Bernoulli array,*

$$\frac{R}{2V} - W_1(S, S^*) = h \sum_{k=0}^{n-1} b_k \operatorname{Var}_{\pi_k}(p_i) = \frac{h}{2} \sum_{k=0}^{n-1} \frac{\sum_{i,j} w_i w_j c_{i,k} c_{j,k} (p_i - p_j)^2}{b_k}. \tag{4.2}$$

Proof. On the sum-lattice cell $[h(k-\mu), h(k+1-\mu)]$, let $s \in (0, 1)$ be the relative position. Condition on the deleted count. The ith replacement in (2.4) gives exactly

$$F_S - F_{T_i + X_i^*} = c_{i,k}(1 - p_i - s).$$

After averaging with the variance weights, this becomes $F_S - F_{S^*} = b_k(Q_k - s)$. The integral of its absolute value on the cell is

$$\frac{hb_k}{2}\{Q_k^2 + (1-Q_k)^2\}.$$

On the other hand, (3.1) gives

$$\frac{R}{2V} = \frac{h}{2}\sum_i w_i\{p_i^2 + (1-p_i)^2\} = \frac{h}{2}\sum_k b_k \mathbb{E}_{\pi_k}\{p_i^2 + (1-p_i)^2\}.$$

Subtracting yields the first identity, since the difference between the two quadratic expressions is $2\,\mathrm{Var}_{\pi_k}(p_i)$. The usual pairwise formula for variance gives the second. □

The weights $\pi_{i,k}$ have a direct probabilistic meaning: conditional on the chosen zero-bias replacement interval having cell index k, they are the probabilities of the replaced coordinate. At $k = 0$ every coordinate has positive conditional weight. Thus (4.2) vanishes exactly when all p_i agree. Conversely, for an iid binary sum all the cell means are $Q_k = 1 - p$, and

$$W_1(S, S^*) = \frac{h}{2}\{p^2 + (1-p)^2\}. \tag{4.3}$$

Together with the strict unequal-diameter argument, this proves Theorem 1.1. Zero coordinates have zero variance weight and do not affect any of the identities.

5 Lattice span, skewness, and transport

We now compare the two terms in Esseen's fixed-law asymptotic formula with the transport distance. Let X be a centered lattice variable of variance $v > 0$ and finite third absolute moment. Suppose its law is supported on $a + h\mathbb{Z}$, with $h > 0$, and write $\kappa = \mathbb{E}X^3$. The spacing h need not be maximal for the inequality below.

Theorem 5.1 (A lattice transport inequality)**.** *For such a law,*

$$|\kappa| + 3hv \le (3 + \sqrt{10})\, 2vW_1(X, X^*). \tag{5.1}$$

Equality holds exactly for the laws of $h(N - np)$, where $N \sim \mathrm{Bin}(n, p)$, $n \ge 1$, and $p \in \{p_{\mathrm{E}}, 1 - p_{\mathrm{E}}\}$.

5.1 Integration on the lattice cells

By (2.1), the density of X^* is constant on every open cell $(a + kh, a + (k+1)h)$; call it d_k. If $d_k = 0$, the cell lies outside the support's convex hull and its CDF difference is zero. For the other cells put

$$\omega_k = hd_k, \qquad r_k = \frac{F_X(a + kh) - F_{X^*}(a + kh)}{hd_k}, \qquad \ell(r) = \int_0^1 |r - s|\, ds.$$

Then $\sum_k \omega_k = 1$. The CDF difference within the cell is $hd_k(r_k - s)$, so

$$\frac{W_1(X, X^*)}{h} = \sum_k \omega_k \ell(r_k), \qquad \sum_k \omega_k (r_k - 1/2) = \frac{\kappa}{2vh}. \tag{5.2}$$

For the second identity, integrate the signed CDF difference and use $\mathbb{E}X^* - \mathbb{E}X = \kappa/(2v)$. Absolute convergence follows from the finite first moments in (2.2).

The function ℓ is convex, with

$$\ell(r) = \begin{cases} r^2 - r + 1/2, & 0 \le r \le 1, \\ |r - 1/2|, & r \notin [0,1]. \end{cases}$$

Let $t = |\kappa|/(vh)$. Jensen's inequality in (5.2) gives

$$2vW_1(X, X^*) \ge \begin{cases} vh(1+t^2)/2, & 0 \le t \le 1, \\ vht, & t \ge 1. \end{cases} \tag{5.3}$$

In the first range, $2(3+t)/(1+t^2)$ has its unique maximum $3+\sqrt{10}$ at $t = \sqrt{10}-3$. In the second range, $(3+t)/t \le 4 < 3+\sqrt{10}$. This proves (5.1).

Combining (5.1) with (1.2) recovers Esseen's lattice moment inequality [3, Lemma 1, pp. 162–163]:

$$|\kappa| + 3hv \le (3+\sqrt{10})\mathbb{E}|X|^3.$$

Esseen's original lemma allows the more general condition that consecutive support points are at least h apart. We use the lattice case because the asymptotic correction is determined by lattice span.

5.2 Equality determines a binomial law

Equality in (5.1) requires $|\kappa|/(vh) = \sqrt{10} - 3 < 1$. The mean argument of ℓ in Jensen's inequality therefore lies strictly between zero and one. At that argument its supporting line touches ℓ only once. All positive-density cells consequently have the same ratio r, with

$$0 < r < 1, \qquad |2r - 1| = \sqrt{10} - 3. \tag{5.4}$$

Let x be a grid point having mass p_x, and let d_- and d_+ be the densities on its left and right cells. The CDF jump and the density jump give

$$p_x = h\{rd_+ + (1-r)d_-\}, \qquad v(d_- - d_+) = xp_x.$$

These equations remain valid at the support endpoints, where one density is zero. Solving them gives

$$hd_- = p_x(1 + rhx/v), \qquad hd_+ = p_x(1 - (1-r)hx/v). \tag{5.5}$$

A missing interior grid atom is impossible: its positive unchanged density would make the cell ratio decrease by one across that point, contrary to its constancy. Positivity in (5.5) confines every atom to $[-v/(rh), v/((1-r)h)]$, so the support is finite. At its lower and upper endpoints,

$$x_- = -\frac{v}{rh}, \qquad x_+ = \frac{v}{(1-r)h}.$$

Their separation is nh for an integer $n \ge 1$. Thus, with $p = 1 - r$,

$$v = nr(1-r)h^2, \qquad x_j = h(j - np), \quad 0 \le j \le n.$$

The density on the cell between x_j and x_{j+1} is the same when calculated from either endpoint. Substituting these locations in (5.5) gives

$$\frac{p_{x_{j+1}}}{p_{x_j}} = \frac{(n-j)p}{(j+1)(1-p)}.$$

Normalization identifies the law as centered $h\,\mathrm{Bin}(n,p)$, and (5.4) gives $p = p_{\mathrm{E}}$ or $1-p_{\mathrm{E}}$. Conversely, (4.3) and the binomial third moment verify equality for each of these laws. This completes the proof of Theorem 5.1.

For $n \geq 2$ these binomial laws are not two-point laws. They attain the span/transport inequality, but their single-law transport bound is strict. Resolving them into their independent binary constituents changes the third-moment denominator and restores equality in (1.4).

6 The Berry–Esseen limit for a fixed block

Return to (1.3). The law of S has finite third absolute moment, since the block is finite. Its variance is V and its signed third moment is

$$\kappa = \mathbb{E}S^3 = \sum_i \mathbb{E}X_i^3.$$

Let h be its maximal lattice span, or set $h = 0$ if its law is nonlattice. Esseen's fixed-law asymptotic formula [3, pp. 160–162, equations (6)–(7)] gives

$$\lim_{m\to\infty} \sqrt{m}\, d_K\left(\frac{S_1 + \cdots + S_m}{\sqrt{mV}}, Z\right) = \frac{\phi(0)}{6V^{3/2}}(|\kappa| + 3hV), \tag{6.1}$$

where Z is standard normal and d_K is Kolmogorov distance. The source uses one fixed law with positive variance and finite third absolute moment. Multiplying by $V^{3/2}/R$ yields

$$\lim_{m\to\infty} C_m = \frac{\phi(0)}{6R}(|\kappa| + 3hV). \tag{6.2}$$

If $h > 0$, Theorem 5.1 applies to the sum law and gives the first inequality in (1.6). Notice that (5.1) has no third absolute moment in its denominator, so it can be divided by the original R without identifying R with $\mathbb{E}|S|^3$. The second inequality is Theorem 1.1.

If $h = 0$, the transport distance dominates the difference of means, so (2.2) gives $|\kappa| \leq 2VW_1(S, S^*)$. Hence the stronger comparison

$$\lim_{m\to\infty} C_m \leq \frac{\phi(0)}{6}\frac{2VW_1(S,S^*)}{R} \leq \frac{\phi(0)}{6} < C_{\mathrm{E}}$$

holds in the nonlattice case.

Finally suppose the limit equals C_{E}. The sum is lattice, and both inequalities in (1.6) are equalities. Theorem 1.1 forces all nonconstant coordinates to have one law $g(B-p)$, with $g > 0$. If there are k such coordinates, the sum is centered $g\,\mathrm{Bin}(k,p)$; its maximal lattice span is g, and

$$\frac{\kappa}{Vg} = 1 - 2p.$$

Equality in Theorem 5.1 forces $|1-2p| = \sqrt{10} - 3$, or $p \in \{p_{\mathrm{E}}, 1-p_{\mathrm{E}}\}$. Conversely these arrays attain both equalities and have the stated limit by (6.1). This proves Theorem 1.2.

References

[1] Logan Bell. Small Lyapunov ratios and the structure of Berry–Esseen maximizers. Companion manuscript, 6 October 2026, 2026. URL https://github.com/epicycloids/independent-berry-esseen-bounds/blob/main/paper/small-lyapunov-structure/small-lyapunov-structure.pdf.

[2] Gustave Choquet. Le théorème de représentation intégrale dans les ensembles convexes compacts. *Annales de l'Institut Fourier*, 10:333–344, 1960. doi: 10.5802/aif.105. URL https://numdam.org/articles/10.5802/aif.105/.

[3] C.-G. Esseen. A moment inequality with an application to the central limit theorem. *Skandinavisk Aktuarietidskrift*, 39:160–170, 1956. doi: 10.1080/03461238.1956.10414946.

[4] Larry Goldstein. Bounds on the constant in the mean central limit theorem. *The Annals of Probability*, 38(4):1672–1689, 2010. doi: 10.1214/10-AOP527. URL https://arxiv.org/abs/0906.5145v2.

[5] Larry Goldstein and Gesine Reinert. Stein's method and the zero bias transformation with application to simple random sampling. *The Annals of Applied Probability*, 7(4):935–952, 1997. doi: 10.1214/aoap/1043862419. URL https://arxiv.org/abs/math/0510619.

[6] Ilya S. Tyurin. New estimates of the convergence rate in the Lyapunov theorem, 2009. URL https://arxiv.org/abs/0912.0726v1. Version 1.